
Online Examination Report

Date:	21.07.2014
Attempts:	1
Examinee:	Naggenda, Geoffrey
Date of birth:	9/23/1990
Subject:	060-Navigation
Result:	54,79 %
Time used:	00:00:12 (HH:MM:SS)

Attachments

1. List of result by question
2. List of result by topic
3. Detailed question result report

Attachment 1: List of result by question

CQB Number	Internal Number	Syllabus reference	Chapter	Points	Max. Points
1	361	61. GENERAL NAVIGATION		1,00	1,00
2	344	61. GENERAL NAVIGATION		0,00	1,00
3	419	61. GENERAL NAVIGATION		1,00	1,00
4	88	61.1.4.4 Isogonals, relation between true and magnetic		1,00	1,00
5	26	61.3.1.1 General properties types projection		0,00	2,00
6	137	61.3.1.3 Lambert conformal conic		0,00	1,00
7	192	61.4.1.2 Heading (compass, magnetic, true, grid)		0,00	1,00
8	90	61.4.1.6 ETA		1,00	1,00
9	187	61.4.2.2 Time		1,00	1,00
10	119	61.4.3 The triangle of velocities, methods of solution fo		2,00	2,00
11	450	62. RADIO NAVIGATION		0,00	1,00
12	452	62. RADIO NAVIGATION		0,00	1,00
13	448	62. RADIO NAVIGATION		0,00	1,00
14	428	62. RADIO NAVIGATION		0,00	1,00
15	374	62. RADIO NAVIGATION		0,00	1,00
16	429	62. RADIO NAVIGATION		1,00	1,00
17	4	62.1 BASIC RADIO PROPAGATION THEORY		1,00	1,00
18	70	62.1.3 Wave propagation		1,00	1,00
19	15	62.1.3.4 Propagation with the frequency bands		1,00	1,00
20	54	62.1.3.4 Propagation with the frequency bands		0,00	1,00
21	331	62.1.3.4 Propagation with the frequency bands		1,00	1,00
22	60	62.1.3.4 Propagation with the frequency bands		0,00	1,00
23	330	62.1.3.4 Propagation with the frequency bands		0,00	1,00
24	94	62.2.2 NDB/ADF		1,00	1,00
25	85	62.2.2.1 Principles		1,00	1,00
26	168	62.2.3 VOR and Doppler-VOR		1,00	1,00
27	129	62.2.3 VOR and Doppler-VOR		0,00	1,00
28	150	62.2.3 VOR and Doppler-VOR		0,00	1,00

CQB Number	Internal Number	Syllabus reference	Chapter	Points	Max. Points
29	147	62.2.3	VOR and Doppler-VOR	1,00	1,00
30	146	62.2.3	VOR and Doppler-VOR	0,00	1,00
31	144	62.2.3	VOR and Doppler-VOR	1,00	1,00
32	297	62.2.3.1	Principles	1,00	1,00
33	131	62.2.3.1	Principles	1,00	1,00
34	133	62.2.3.2	Presentation and interpretation	0,00	1,00
35	154	62.2.3.3	Coverage and Range	0,00	1,00
36	294	62.2.3.3	Coverage and Range	1,00	1,00
37	143	62.2.3.3	Coverage and Range	1,00	1,00
38	242	62.2.4	DME	0,00	1,00
39	292	62.3.1	Pulse techniques and associated terms	1,00	1,00
40	234	62.3.3.2	Presentation and interpretation	1,00	1,00
Total:55%				23,00	42,00

Attachment 2: List of result by topic

Licence

Date: 17.07.2014

Examinee: Naggenda, Geoffrey

Location: Uganda Civil Aviation Authority

Your Result: 23,00 from 42,00 Points (54,76 %)

Topics	Amount of Questions	Points achieved	Maximum Points	Percent
61. GENERAL NAVIGATION	10	7,00	12,00	58,33
61.1. BASICS OF NAVIGATION	1	1,00	1,00	100,00
61.3. CHARTS	2	0,00	3,00	0,00
61.4. DEAD RECKONING NAVIGATION (DR)	4	4,00	5,00	80,00
62. RADIO NAVIGATION	30	16,00	30,00	53,33
62.1. BASIC RADIO PROPAGATION THEORY	7	4,00	7,00	57,14
62.2. RADIO AIDS	15	9,00	15,00	60,00
62.3. RADAR	2	2,00	2,00	100,00
Total	40	23,00	42,00	54,76

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1

(Refer to Figure 40.) The line from point C to point A of the wind triangle represents

See Attachments:

1. cfi_40.jpg

figure 40

(Attachment
No.1)

☐ true course and groundspeed.

☒ wind direction and velocity.

☐ true heading and groundspeed.

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2

How far will an aircraft travel in 2,1/2 minutes with a groundspeed of 98 knots?

☐ 4.08 NM.

☐ 3.35 NM.

☒ 2.45 NM.

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3

Which of the following defines the type of altitude used when maintaining FL070?

☐ Calibrated.

☒ Pressure.

☐ Indicated.

4

Isogonals are lines of equal:

- ☐ wind velocity.
- ☒ magnetic variation.
- ☐ pressure.
- ☐ compass deviation.

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5

A chart has the scale 1: 1 000 000. From A to B on the chart measures 3.8 cm, the distance from A to B in NM is:

☐ 20.5

☒ 205

☐ 70.4

☐ 38

6

A Lambert conformal conic projection, with two standard parallels:

- ☐ shows all great circles as straight lines
- ☐ shows lines of longitude as parallel straight lines
- ☒ the scale is only correct at parallel of origin
- ☐ the scale is only correct along the standard parallels

7

Given:
True Track 245°
Drift 5° right
Variation 3° E
Compass Hdg 242°

Calculate the deviation.

☐ 1° E

☐ 11° E

☒ 5° E

☐ 5° W

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8

Given: $GS = 120 \text{ kt}$.
Distance from A to B = 84 NM.

What is the time from A to B?

☐ 00 HR 44 MIN

☐ 00 HR 43 MIN

☒ 00 HR 42 MIN

☐ 00 HR 45 MIN

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9

How long will it take to fly 5 NM at a groundspeed of 269 Kt ?

☐ 1 MIN 55 SEC

☐ 0 MIN 34 SEC

☐ 2 MIN 30 SEC

☒ 1 MIN 07 SEC

10

Given:

True Track = 095° ,

TAS = 160 kt,

True Heading = 087° ,

GS = 130 kts;

Calculate W/V

☐ **$237^\circ/36$ kt**

☐ **$124^\circ/36$ kt**

☒ **$057^\circ/36$ kt**

☐ **$307^\circ/36$ kt**

11

Which of the following should be considered as station passage when using VOR?

- ☐ The first full-scale deflection of the CDI.
- ☐ The first complete reversal of the TO-FROM indicator.
- ☒ The first flickering of the TO-FROM indicator and CDI as the station is approached.

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12

A VOR receiver with normal five-dot course sensitivity shows a three-dot deflection at 30 NM from the station. The aircraft would be displaced approximately how far from the course centerline?

☐ 5 NM.

☐ 3 NM.

☒ 2 NM.

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What angular deviation from a VOR course centerline is represented by a full-scale deflection of the CDI?

☐ 4degrees.

☐ 10degrees.

☒ 5 degrees.

14

Where does the DME indicator have the greatest error between ground distance to the VORTAC and displayed distance?

- ☐ High altitudes far from the VORTAC.
- ☒ Low altitudes far from the VORTAC.
- ☐ High altitudes close to the VORTAC.

15

When is the course deviation indicator (CDI) considered to have a full-scale deflection?

- ☐ When the CDI deflects from half-scale left to half-scale right, or vice versa.
- ☐ When the CDI deflects from the center of the scale to full-scale left or right.
- ☒ When the CDI deflects from full-scale left to full-scale right, or vice versa.

16

What indication should a pilot receive when a VOR station is undergoing maintenance and may be considered unreliable?

- ☐ A voice recording on the VOR frequency announcing that the VOR is out of service for maintenance.
- ☐ Coded identification, but no navigation indications.
- ☒ No coded identification, but possible navigation indications.

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17

The transmission frequency is 85 MHz, what is the wavelength?

☐ 2.83 m

☐ 2.55 m

☒ 3.53 m

18

The most common source of static is:

☐ Noise.

☒ Lightning.

☐ Turbulence

19

what is Frequency?

- ☐ The distance travelled during the transmission in one cycle.
- ☒ The number of cycles per second.
- ☐ One complete change of direction of current.

20

The international distress frequencies are:

☐ 500 kHz 121.5 kHz 243 MHz 8364 MHz

☐ 500 kHz 121.5 MHz 243 MHz 8364 kHz

☒ 500 MHz 121.5 MHz 243 KHZ 8364 MHz

21

A carrier wave with amplitude of 5V is modulated by an audio frequency with amplitude of 3 V. The depth of modulation is:

- ☒ 60 %;
- ☐ 40 %;
- ☐ 167 %.

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22

The "squench" control on a VHF communication receiver:

- ☒ Prevents feed-back.
- ☐ Extends the range of the receiver.
- ☐ Disables receiver output when no signals are being received so preventing noise being fed to crew headsets.

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23

The frequency band in which Secondary Surveillance Radar (SSR) operates is:

☒ VHF;

☐ SHF.

☐ UHF;

24

Which four factors affect the accuracy of an NDB?

- ☐ Quadrantal error, night effect, site error, interference.
- ☐ Quadrantal error, night effect, pilotage error, propagation error.
- ☒ Quadrantal error, mountain effect, thunderstorm effect, night effect.

25

The ADF pointer measures a bearing from an NDB relative to:

☒ Aircraft heading.

☐ Magnetic north.

☐ Aircraft track.

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26

From where can ATIS information be received?

☒ **At certain times from airfield VORs.**

☐ **At all times from any VOR.**

☐ **From an ILS transmission.**

27

A VOR CDI has the OBS set at 090. The variable phase signal lags the reference signal by 280. The CDI indication will be:

- ☒ **Needle LEFT with TO indicated.**
- ☐ **Needle Right with FROM indicated.**
- ☐ **Needle RIGHT with TO indicated.**

28

A full scale deflection on a VOR receiver occurs when the aircraft is:

- ☐ 5 or more from the selected bearing.
- ☒ 1 from the selected course when close to the station and 10 from the selected course when far from the station.
- ☐ 10 or more from the selected bearing.

29

After passing overhead a VOR the CDI shows a scale deviation to the right. What is indicated if the deflection remains constant for a period of time?

- ☐ The aircraft is flying closer to the radial.
- ☒ The aircraft is flying further from the radial.
- ☐ The OBS is erroneously set on the reciprocal heading.

30

When the VOR indicates TO and the CDI is centered, where will the aircraft always be?

- ☒ On the radial identified on the course selector.
- ☐ On the reciprocal of the radial identified on the course selector.
- ☐ Inbound to the VOR.

31

Which of the following factors would affect the range of the VOR?

- ☐ Transmission power, transmitter/receiver height, designated operational coverage, night effect.
- ☐ Transmission power, frequency in use, type of emission, receiver/transmitter height, nature of terrain.
- ☒ Designated operational coverage, transmitter/receiver height, transmission power, nature of terrain.

32

The principle of VOR is bearing measurement by:

- ☒ **phase comparison;**
- ☐ **limaçon positions.**
- ☐ **wave transmissions;**

33

The principle of operation of a VOR is bearing measurement by:

☐ Modulation measurement.

☒ Phase comparison.

☐ Polarisation comparison.

34

An aircraft is flying on a heading of 270°(M). The VOR OBS is also set to 270° with the full left deflection and FROM flag displayed.

In which sector is the aircraft from the VOR ground station?

☐ NE

☐ NW

☒ SE

☐ SW

35

With reference to the VOR when the aircraft heading agrees approximately with the OBS setting:

- ☐ Fly towards the CDI provided FROM is indicated.
- ☒ Fly away from the needle provided FROM is indicated.
- ☐ Fly towards the CDI regardless of the TO/FROM indication.

36

The factors affecting the accuracy of a VOR reading are:

- ☒ airborne equipment; site error; propagation error; interference error.
- ☐ airborne equipment; site error; conduction error; propagation error.
- ☐ airborne equipment; site error; propagation error; refraction error.

37

Which of the following factors would affect the accuracy of a VOR?

- ☐ Propagation error, site error, pilotage error, transmitting power.
- ☐ Site error, propagation error, night effect, airborne equipment error.
- ☒ Site error, propagation error, airborne equipment error, interference error.

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38

What is the DME frequency band?

☒ VHF

☐ UHF

☐ VLF

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39

Basic radar operates on:

- ☒ VHF and higher frequencies;
- ☐ Any frequency in the RF band.
- ☐ HF and lower frequencies only;

40

The PRF of a radar installation is 570 pps. What is the time interval between successive transmissions of the leading edges of the pulse?

☐ 1643.85 s.

☐ 1140.00 s.

☒ 1754.39 s.

Attachment No. 1

Questions 1

figure 40

cfi_40.jpg

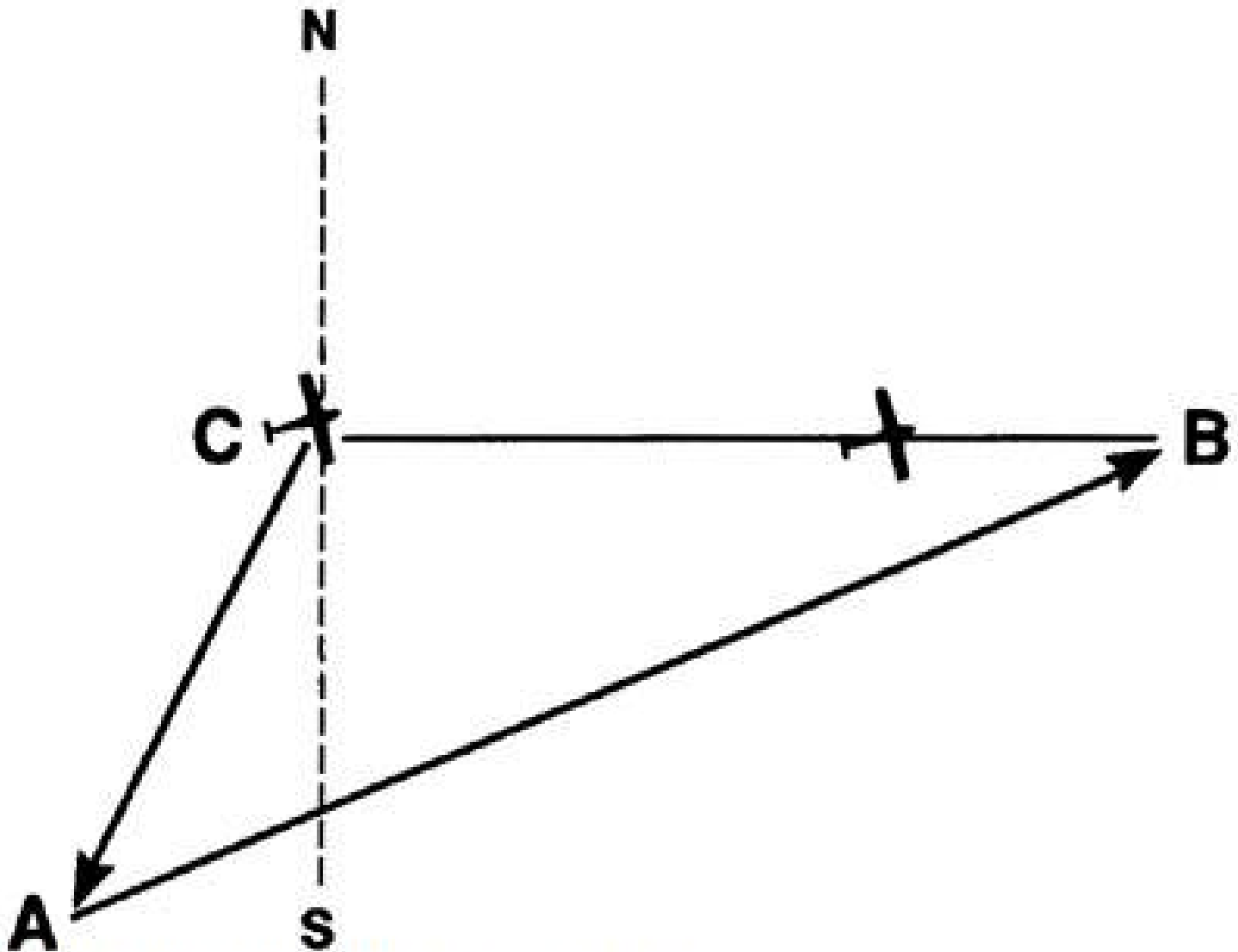


Figure 40. Wind Triangle © ASA

Comment No.2 for question 12:

12 not for ppl